

Toru Takisaka

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Academic Appointments and Education

- NOV. 2023–PRESENT Associate Professor at University of Electronic Science and Technology of China, China
Algorithms and Logic Group
- APR. 2023–OCT. 2023 Research Associate Professor at University of Electronic Science and Technology of China, China
Algorithms and Logic Group
- APR. 2022–MAR. 2023 Postdoctoral Researcher at University of Electronic Science and Technology of China, China
Algorithms and Logic Group
Supervisor: Bakhadyr Khoussainov
- APR. 2018–MAR. 2022 Postdoctoral Researcher at National Institute of Informatics (NII), Japan
ERATO Metamathematics for Systems Design Project
Supervisor: Ichiro Hasuo
- APR. 2014–MAR. 2018 Ph.D. student at Research Institute for Mathematical Sciences, Kyoto University, Japan
Dissertation title: Large Scale Geometries of Infinite Strings
Supervisor: Masahito Hasegawa
- APR. 2012–MAR. 2014 M.Sc. student at Research Institute for Mathematical Sciences, Kyoto University, Japan
Thesis title: On Gács' Quantum Algorithmic Entropy
Supervisor: Masahito Hasegawa

Publications

Journal articles

1. Sasinee Pruekprasert, Jérémy Dubut, **Toru Takisaka**, Clovis Eberhart, and Ahmet Cetinkaya: *Moment propagation of polynomial systems through Carleman linearization for probabilistic safety analysis*. *Automatica*, 160: 111441, 2024.
2. **Toru Takisaka**, Zhenya Zhang, Paolo Arcaini, and Ichiro Hasuo: *A Robustness-Based Confidence Measure for Hybrid System Falsification*. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 42(5), pp. 1718–1731, 2023. [CCF-A journal]
3. Bakhadyr Khoussainov and **Toru Takisaka**: *Infinite Strings and Their Large Scale Properties*. *The Journal of Symbolic Logic (JSL)*, 87(2), pp. 585–625, 2022.
4. **Toru Takisaka**, Yuichiro Oyabu, Natsuki Urabe and Ichiro Hasuo: *Ranking and Repulsing Supermartingales for Reachability in Randomized Programs*. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 43(2), Article 5, pp. 1–46, 2021. [CCF-A journal]

Conference proceedings

5. Libo Zhang, Yang Chen, **Toru Takisaka**, Kaiqi Zhao, Weidong Li, and Jiamou Liu: *Situational-Constrained Sequential Resources Allocation via Reinforcement Learning*. Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), pp. 9121–9129, 2025. [CCF-A, CORE-A* conference]
6. **Toru Takisaka**, Libo Zhang, Changjiang Wang, and Jiamou Liu: *Lexicographic Ranking Supermartingales with Lazy Lower Bounds*. Proceedings of the 36th International Conference on Computer-Aided Verification (CAV), LNCS 14683, pp. 420–442, 2024. [CCF-A, CORE-A* conference]
7. Zihui Liang, Bakhadyr Khoussainov, **Toru Takisaka**, and Mingyu Xiao: *Connectivity in the presence of an opponent*. Proceedings of the 31st Annual European Symposium on Algorithms (ESA), LIPIcs 274, pp. 79:1–79:14, 2023. [CCF-B, CORE-A conference]
8. Libo Zhang, Yang Chen, **Toru Takisaka**, Bakhadyr Khoussainov, Michael Witbrock, and Jiamou Liu: *Learning Density-Based Correlated Equilibria for Markov Games*. Proceedings of the 22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS), pp. 652–660, 2023. [CCF-B, CORE-A* conference]
9. Ezequiel Castellano, Xiao-Yi Zhang, Paolo Arcaini, **Toru Takisaka**, Fuyuki Ishikawa, Nozomu Ikehata, and Kosuke Iwakura: *Explaining the Behaviour of Game Agents Using Differential Comparison*. Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (ASE; ASE4Games 2022 workshop), Article 210, pp. 1–8, 2022.
10. Masaki Waga, Ezequiel Castellano, Sasinee Pruekprasert, Stefan Klikovits, **Toru Takisaka**, and Ichiro Hasuo: *Dynamic Shielding for Reinforcement Learning in Black-Box Environments*. Proceedings of the 20th International Symposium on Automated Technology for Verification and Analysis (ATVA), LNCS 13505, pp. 25–41, 2022.
11. Kittiphon Phalakarn, **Toru Takisaka**, Thomas Haas and Ichiro Hasuo: *Widest Paths and Global Propagation in Bounded Value Iteration for Stochastic Games*. Proceedings of the 32nd International Conference on Computer-Aided Verification (CAV), LNCS 12225, pp. 349–371, 2020. [CCF-A, CORE-A* conference]
12. Sasinee Pruekprasert, **Toru Takisaka**, Clovis Eberhart, Ahmet Cetinkaya and Jérémy Dubut: *Moment Propagation of Discrete-Time Stochastic Polynomial Systems using Truncated Carleman Linearization*. IFAC-PapersOnLine 53(2), pp. 14462–14469, 2020.
13. **Toru Takisaka**, Yuichiro Oyabu, Natsuki Urabe and Ichiro Hasuo: *Ranking and Repulsing Supermartingales for Reachability in Probabilistic Programs*. Proceedings of the 16th International Symposium on Automated Technology for Verification and Analysis (ATVA), LNCS 11138, pp. 476–493, 2018.
14. Bakhadyr Khoussainov and **Toru Takisaka**: *Large Scale Geometries of Infinite Strings*. Proceedings of the 32nd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS), pp. 1–12, 2017. [CCF-A, CORE-A* conference]
15. **Toru Takisaka**: *On Gács’ Quantum Algorithmic Entropy*. Proceedings of the 11th Workshop on Quantum Physics and Logic (QPL), EPTCS 172, pp. 204–216, 2014.

Research Grants

16. Research Fund for International Young Scientists
National Natural Science Foundation of China, Grant number 62150410437, Jan. 2022–Dec. 2022.
17. Foreign Experts Key Support Program Project – Foreign Scholars Research Fund Project
Ministry of Science and Technology of the People’s Republic of China, Jan. 2022–Dec. 2022.

Honors and Awards

18. Recipient, Excellent Young Scientists Fund (Overseas)
National Natural Science Foundation of China, 2023.

Invited Talks

19. **Toru Takisaka:** *Large Scale Geometries of Infinite Strings*. Oral presentation delivered at the 16th Asian Logic Conference (ALC), Astana, Kazakhstan, June 2019.
20. **Toru Takisaka:** *Ranking and Repulsing Supermartingales for Reachability in Probabilistic Programs*. Oral presentation delivered at the 35th Japan Society for Software Science and Technology (JSSST) Annual Conference, Osaka, Japan, August 2018.

Patents

21. U.S. Patent No. 12,360,519, "System optimal control device, system optimal control method, and program", granted on July 15, 2025.
Inventors: Kittiphon Phalakarn, **Toru Takisaka**, Thomas Haas and Ichiro Hasuo.
22. Japanese Patent No. 7636652, "System optimal control device, system optimal control method, and program", granted on February 18, 2025.
Inventors: Kittiphon Phalakarn, **Toru Takisaka**, Thomas Haas and Ichiro Hasuo.

Teaching Experience

University of Electronic Science and Technology of China, China

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| FALL 2026–PRESENT | Instructor, Formal Methods (Graduate course) |
| SPRING 2024–PRESENT | Instructor, Mathematical Foundation of Computer Science (Undergraduate course) |
| FALL 2025 | Instructor, Cloud Computing (Graduate course) |

National Institute of Informatics (NII), Japan

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| SPRING 2019 | Guest lecturer, Formal Methods for Cyber-Physical Systems |
| FALL 2018 | Teaching assistant, Formal Language Theory |

Professional Service

Program committee member

- International Computing and Combinatorics Conference (COCOON 2025)
JSSST Workshop on Programming and Programming Languages (PPL 2021, PPL 2024)

Organizing committee member

- International Computing and Combinatorics Conference (COCOON 2025)
International Conference and Workshops on Algorithms and Computation (WALCOM 2025)

External reviewer (selected)

- Conferences:** ITP 2026, CONCUR 2025, MFCS 2024, ICALP 2024, COCOON 2024, ATVA 2023, ISAAC 2023, ESOP 2021, CONCUR 2019, LICS 2018
Journals: Logical Methods in Computer Science (LMCS), Formal Aspects of Computing (FAC)